Capital University [*formerly* Helwan University]

Mainstream Program – Semester 2 [Spring 2026]

Module: **CS251 Software Engineering – 1**

[**Group-Project Technical Requirements** | *Version* **1.0**]

Project Management:

Group Size	• 6 to 7 members
Deliveries	• Phase 1: Week 9 (4-4-2026) • Phase 2: Week 14 (9-5-2026)

Tools:

Visual Paradigm (Community)	Visual modeling tool used primarily for system design and architecture • https://www.visual-paradigm.com/download/community.jsp
Xampp	XAMPP is a software bundle that turns your computer into a local server for building and testing PHP web applications • https://www.apachefriends.org/download.html
VS Code (IDE)	Visual Studio Code is a highly customizable code editor focused on speed and developer experience • https://code.visualstudio.com/
	Extensions • PHP Intelephense • Live Server
Database	MySQL (managed via phpMyAdmin) • *Note: While MySQL is standard, any database engine is permitted*

Approved Development Stack:

Language	• PHP
Framework	• PHP Native • *Note: The Laravel Framework is also permitted but is not mandatory.*
Front-end Libraries	• Bootstrap or Tailwind CSS
Front-end Templates	• Prohibited (Ready-made templates are not allowed)

Technical Requirements Required:

Architecture (Design Patterns)	• MVC Architecture (Model-View-Controller) • Object-Oriented Programming (OOP) • Singleton Design Pattern for Database Connection
Database Connectivity	• MySQL (via phpMyAdmin) with MySQLi or PDO
Security & Auth	• Authentication using Password Hashing • Role-based Authorization • PHP Sessions
Data Management	• PHP Implementation of **CRUD operations** (Create, Read, Update, Delete)
Features	• File Uploading • AJAX Calls (in one scenario)

Documentation & Design (UML)

Software Requirement Specification (SRS)	• Functional (Use Case Scenarios) and Non-Functional Requirements (*for each, both User and System Requirements*) • Use Case Diagram + Descriptions • Activity Diagrams / Swimlane Diagrams
Software Design Document (SDD)	• Architectural Design (a block diagram) • Class Diagrams • Sequence Diagrams • Entity Relationship Diagrams (ERD) • Object Diagrams • Communications/Collaboration Diagrams • Package Diagrams